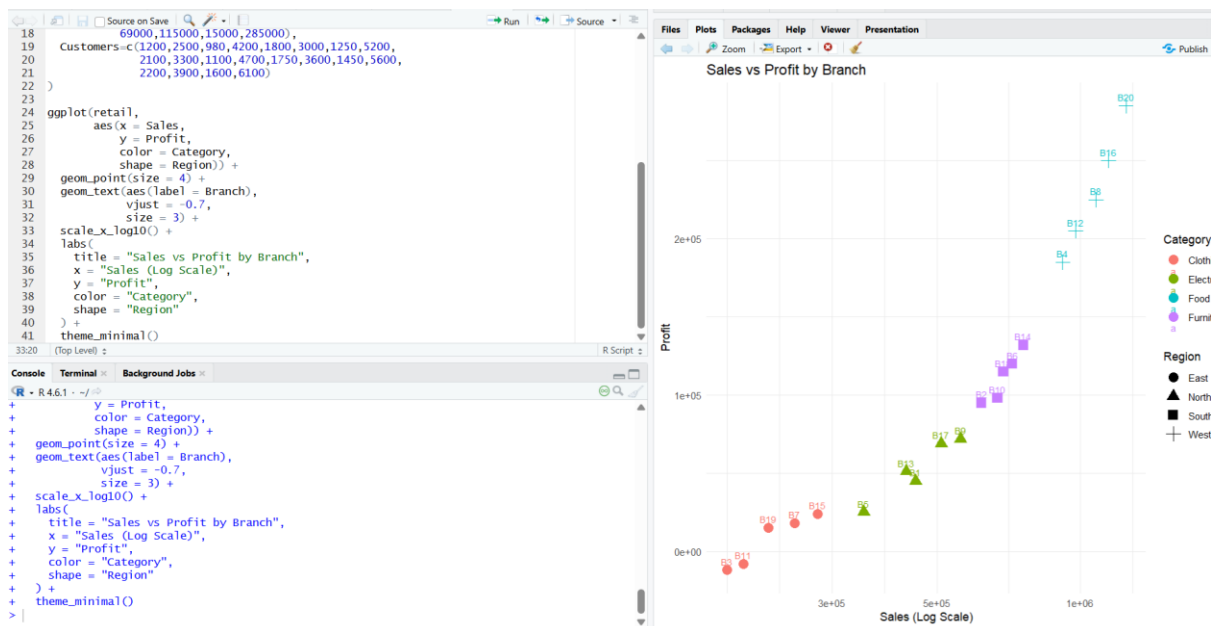


ASSESSMENT TOOL 3

Name : Jayaprakash Nivas

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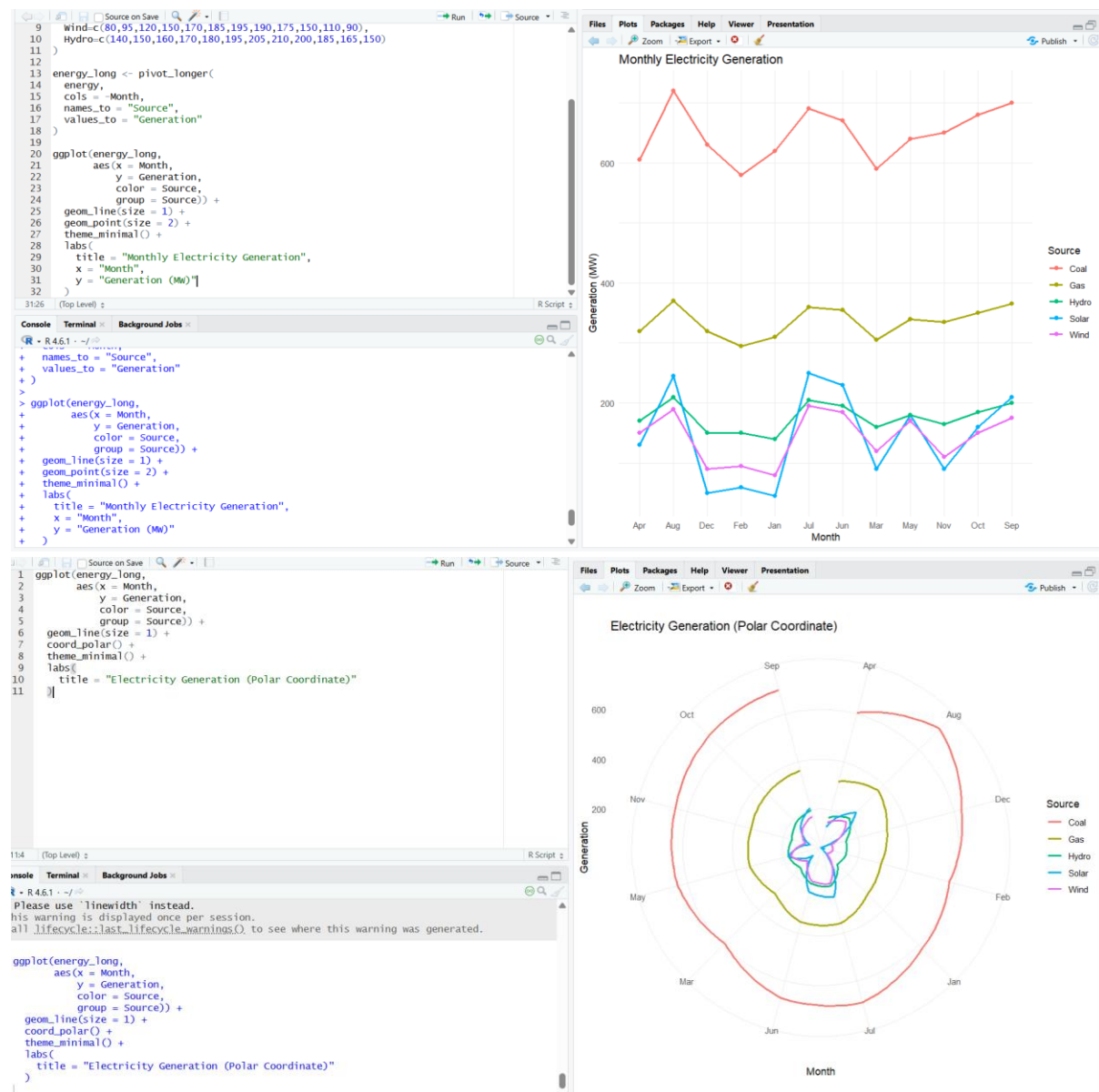
Question 1



Explanation :

- Scatter Plot is used to show the relationship between Sales and Profit.
- Sales is mapped to the X-axis for comparison across branches.
- Profit is mapped to the Y-axis to analyze profitability.
- Color represents different Product Categories (Electronics, Furniture, Clothing, Food).
- Shape represents different Regions (North, South, East, West).
- Branch names are added as labels to identify each data point.
- `scale_x_log10()` is applied to the Sales axis because sales values have a wide range, improving readability.
- `theme_minimal()` is used to provide a clean and professional appearance.

Question 2



Explanation (Cartesian Coordinate - Line Chart)

- Line Chart is used to show the monthly trend in electricity generation.
- Month is mapped to the X-axis.
- Electricity Generation (MW) is mapped to the Y-axis.
- Color represents different energy sources (Coal, Gas, Solar, Wind, and Hydro).
- Lines connect monthly values to show increasing or decreasing trends.
- `theme_minimal()` is used to provide a clean and professional appearance.

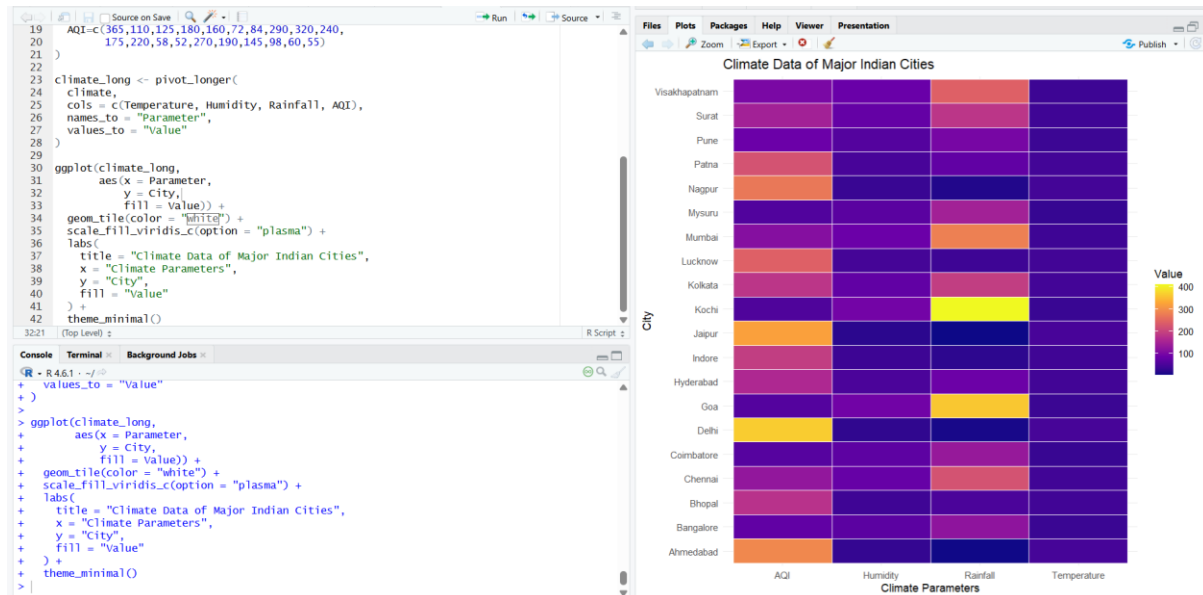
Explanation (Polar Coordinate Chart)

- Polar Coordinate Chart represents the same data in a circular layout.
- Months are arranged around the circle.
- Distance from the center represents electricity generation.
- Color distinguishes different energy sources.
- `coord_polar()` converts the line chart into a circular (polar) chart.
- The circular layout helps visualize seasonal or cyclical patterns throughout the year.

Comparison

- Cartesian Coordinate is better for comparing exact values and identifying trends over time.
- Polar Coordinate is useful for highlighting cyclical or seasonal patterns.
- Cartesian Coordinate provides better readability and accurate comparison, while Polar Coordinate offers a more visual representation of periodic data.

Question 3



Explanation

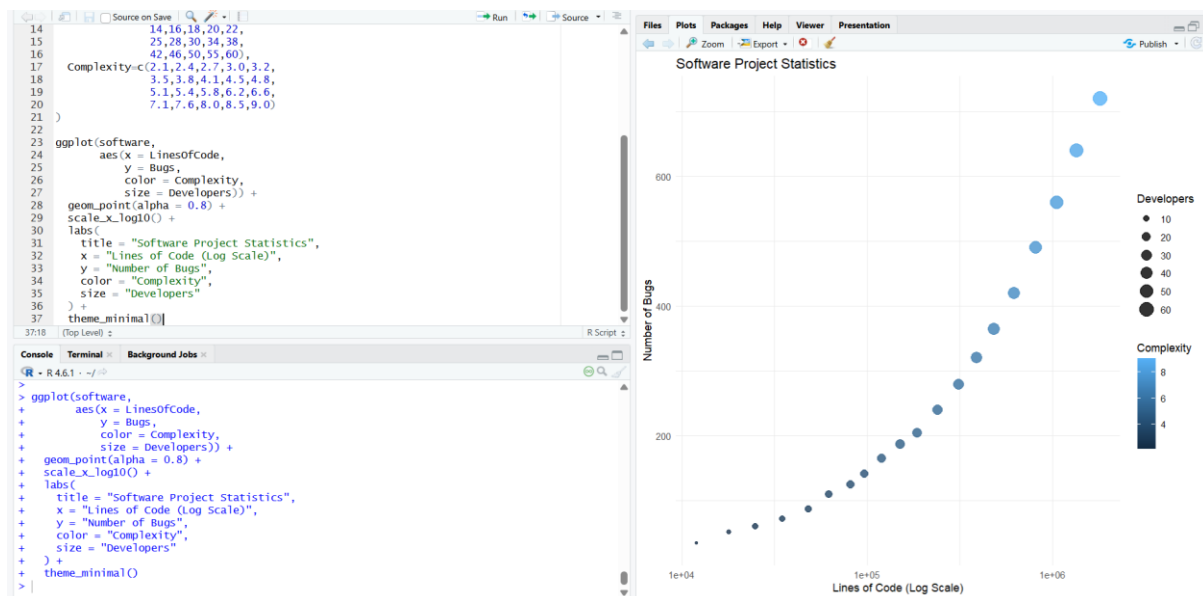
- Heatmap is used to compare multiple climate parameters across different cities.
- Climate Parameters (Temperature, Humidity, Rainfall, AQI) are mapped to the X-axis.
- Cities are mapped to the Y-axis.

- Fill colour represents the magnitude of each climate parameter.
- `scale_fill_viridis_c(option = "plasma")` applies the Plasma colour palette.
- The Plasma palette provides high contrast, making low and high values easy to distinguish.
- It is colour-blind friendly, ensuring accessibility for all users.
- `theme_minimal()` gives the visualization a clean and professional appearance.

Justification of Colour Palette

- Plasma offers a smooth gradient from low to high values.
- It improves readability and accurately highlights variations in climate data.
- It is perceptually uniform and suitable for users with colour vision deficiencies, making the visualization more accessible.

Question 4



Explanation

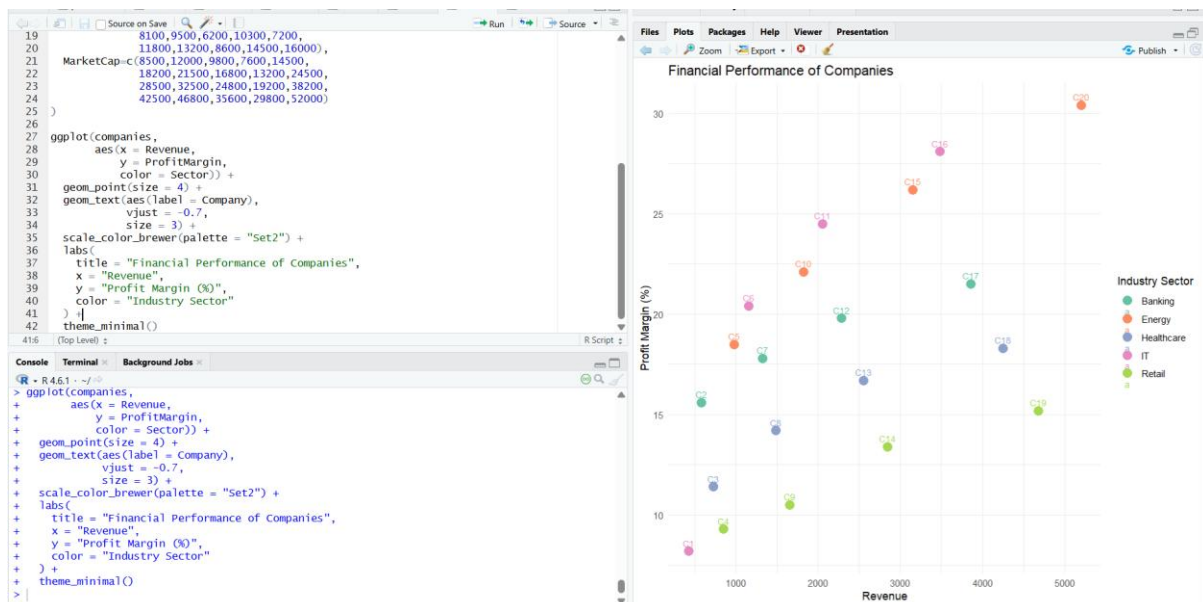
- Scatter Plot is used to show the relationship between Lines of Code and Number of Bugs.
- Lines of Code is mapped to the X-axis.
- Number of Bugs is mapped to the Y-axis.
- Colour represents the Complexity of the software project.
- Point size represents the Number of Developers.
- `scale_x_log10()` applies a logarithmic (nonlinear) transformation to the X-axis.

- The logarithmic scale compresses very large values while preserving the relationship between observations.
- `theme_minimal()` provides a clean and professional appearance.

Justification of the Transformation

- The Lines of Code values range from 12,000 to 1,800,000, making the data highly skewed.
- A linear scale would cluster smaller values together and make comparisons difficult.
- A logarithmic scale (`scale_x_log10()`) spreads the smaller values and compresses the larger ones, improving readability and making trends easier to interpret.

Question 5



Explanation

- Scatter Plot is used to show the relationship between Revenue and Profit Margin.
- Revenue is mapped to the X-axis.
- Profit Margin is mapped to the Y-axis.
- Industry Sector is represented using different colours.
- Company names are displayed as labels to identify each data point.
- Cartesian coordinate system is used for accurate comparison of financial performance.
- `scale_color_brewer(palette = "Set2")` applies a colour-blind-friendly palette, improving accessibility and reducing perceptual bias.
- `theme_minimal()` provides a clean and professional appearance.

Justification

- The scatter plot clearly shows the relationship between Revenue and Profit Margin.
- Different colours make it easy to distinguish companies from different Industry Sectors.
- The Set2 colour palette provides good contrast and is suitable for accessibility.
- The Cartesian coordinate system allows accurate comparison between companies.